Unsolved Exercises on Truth Tables

Exercise 1

Complete the truth table for:

$$p \wedge q$$

p	**q**	**p ∧ q**
T	T	___
T	F	___
F	T	___
F	F	___

Exercise 2

Complete the truth table for:

$$p \vee q$$

p	**q**	**p ∨ q**
T	T	___
T	F	___
F	T	___
F	F	___

Exercise 3

Complete the truth table for:

$$p \rightarrow q$$

p	**q**	**p → q**
T	T	___
T	F	___
F	T	___
F	F	___

Exercise 4

Complete the truth table for:

$$p \leftrightarrow q$$

p	**q**	**p ↔ q**
T	T	___
T	F	___
F	T	___
F	F	___

Exercise 5

Complete the truth table for:

$$\neg p \vee q$$

p	q	¬p	¬p ∨ q
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 6

Complete the truth table for:

$$p \wedge \neg q$$

p	q	¬q	p ∧ ¬q
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 7

Complete the truth table for:

$$\neg(p \wedge q)$$

p	q	p ∧ q	¬(p ∧ q)
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 8

Complete the truth table for:

$$\neg(p \vee q)$$

p	q	p ∨ q	¬(p ∨ q)
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 9

Complete the truth table for:

$$(p \wedge q) \rightarrow p$$

p	q	p ∧ q	(p ∧ q) → p
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 10

Complete the truth table for:

$$p \to (q \vee p)$$

p	q	q ∨ p	p → (q ∨ p)
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 11

Complete the truth table for:

$$(p \to q) \wedge p$$

p	q	p → q	(p → q) ∧ p
T	T	___	___
T	F	___	___
F	T	___	___
F	F	___	___

Exercise 12

Complete the truth table for:

$$(p \vee q) \wedge \neg p$$

p	q	p ∨ q	¬p	(p ∨ q) ∧ ¬p
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

Exercise 13

Complete the truth table for:

$$(p \wedge q) \vee \neg q$$

p	q	p ∧ q	¬q	(p ∧ q) ∨ ¬q
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___

Exercise 14

Complete the truth table for:

$$(p \to q) \leftrightarrow (\neg q \to \neg p)$$

p	q	p → q	¬q	¬p	¬q → ¬p	Final
T	T	___	___	___	___	___
T	F	___	___	___	___	___
F	T	___	___	___	___	___
F	F	___	___	___	___	___

Exercise 15

Complete the truth table for:

$$(p \vee q) \rightarrow (p \wedge q)$$

p	**q**	**p ∨ q**	**p ∧ q**	**(p ∨ q) → (p ∧ q)**
T	T	___	___	___
T	F	___	___	___
F	T	___	___	___
F	F	___	___	___